

GeoCoT: Towards Reliable Remote Sensing Reasoning with Manifold Perspective

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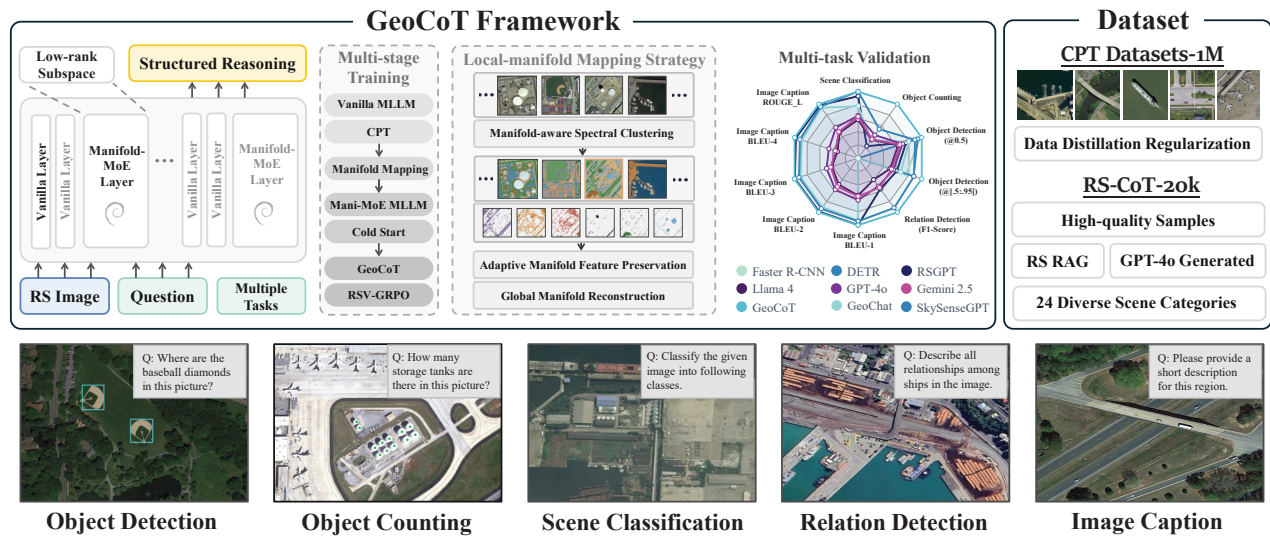


Figure 1. Overview of GeoCoT and RS-CoT-20k dataset applied to multi-task scenes.

Abstract

Multimodal Large Language Models (MLLMs) have shown strong potential in remote sensing (RS) through multi-task reasoning and cross-modal generalization. However, existing RS-MLLMs mainly rely on a single shared expert for all tasks, making it hard to produce reliable results. Meanwhile, the intrinsic redundancy and homogeneity of RS images bring substantial difficulties for both training and inference. These challenges directly conflict with the demands of remote sensing, which values task precision and trustworthy reasoning. To address these limitations, we propose GeoCoT, a manifold-driven Mixture-of-Experts (MoE) system with Chain-of-Thought (CoT) reasoning. GeoCoT introduces Mani-MoE, a sparse expert architecture grounded

in local manifold mapping. It projects high-dimensional tokens onto low-rank subspaces adaptively to eliminate redundancy and uncover intrinsic structure, and then routes them through a sparse expert pathway, where gating decisions are guided by the manifold structure of the input. To optimize this architecture, we adopt a CoT-driven multi-stage training strategy. It leverages a cold-start phase for domain adaptation, followed by our RS Vision Group Relative Policy Optimization (RSV-GRPO) to systematically strengthen structured reasoning from global to objectives. Furthermore, we innovatively build RS-CoT-20k dataset for task-specific supervision. Extensive experiments across diverse multi-task RS benchmarks demonstrate that GeoCoT outperforms prior approaches, achieving 5.27% higher average accuracy than the state-of-the-art method.

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1. Introduction

Remote sensing (RS) image understanding plays a vital role in geographic applications such as land cover classification and object detection [7, 11, 17, 47]. Traditional single-task RS models [20, 27, 56] usually focus on a narrow scene and struggle to meet the requirements of different downstream tasks simultaneously, lacking the generalization capabilities required for real-world geospatial analysis [6, 51]. To address this limitation, Remote Sensing Multimodal Large Language Models (RS-MLLMs) [26, 28] have emerged as a promising paradigm to align vision and language modalities for multi-task reasoning. Despite this progress, current RS-MLLMs still lack specially structured and highly reliable outputs when providing RS scene reasoning [12, 45]. Since existing methods rely on a single shared parameter set to implement RS tasks in multiple domains, and this dense architecture [30, 50] fails to deliver the fine-grained task adaptability needed for complex RS scenes [37, 39, 46, 57].

To solve this challenge, an effective solution is to use the Mixture-of-Experts (MoE) [10, 13, 24], which aims to achieve the optimal model in multiple tasks by setting multiple weight subspaces. However, compared with the MoE method used for natural images, due to the large amount of redundant information and noise in RS images, direct data routing will introduce substantial redundant computational overhead and reduce the model performance [2, 44, 48, 49]. Meanwhile, existing RS-MLLMs are often deficient in structured reasoning capabilities [16, 43, 55]. This leads to reasoning results that lack a coherent logic from global scene understanding to task-specific goal interpretation, thus undermining reliability in complex RS scenes.

To overcome these limitations, we propose GeoCoT, as shown in Figure 1, a manifold-driven MoE system with Chain-of-Thought (CoT) reasoning. Different from current MoE methods, the proposed Mani-MoE maps the high-dimensional RS data to a low-rank manifold subspace that captures the intrinsic salient features, mitigating unnecessary focus as well as noise interference while enhancing task-level accuracy. Especially, to align model behavior with structured reasoning, we introduce a multi-stage CoT-driven training strategy. After continual pre-training and cold start, RS Vision Group Relative Policy Optimization (RSV-GRPO) is leveraged to guide structured and high-precision reasoning of the model through a rule-based reward function. Furthermore, we construct *RS-CoT-20k*, a task-diverse dataset with 20k annotated samples. Extensive experiments on multi-task datasets show that GeoCoT outperforms state-of-the-art methods across tasks.

Our contributions in this paper are highlighted as follows: (1) We propose a novel manifold-driven MoE framework that maps high-dimensional RS data to a low-rank manifold subspace to reduce the redundant calculation of tokens as well as noise interference, while dynamically allo-

cating specialized experts for improved performance and realizing task adaptation. (2) We develop a multi-stage training strategy driven by our RSV-GRPO mechanism to guide the model in structured and high-precision reasoning. (3) We construct *RS-CoT-20k*, a new dataset of multi-modal remote sensing examples with diverse tasks and expert-crafted CoT annotations. This dataset supports supervised training and rigorous evaluation of complex reasoning in RS-MLLMs. (4) Extensive ablation and comparison experiments demonstrate that GeoCoT achieves superior accuracy and generalization across diverse remote sensing tasks.

2. Related Work

Remote sensing encompasses a broad range of tasks [9, 21, 32], all requiring multi-scale visual reasoning grounded in geospatial semantics. Early efforts [4, 19, 29, 35, 42] focus on a single task and lack the generalization across such heterogeneous demands, making RS-MLLMs increasingly vital for unified and scalable RS understanding. Recent RS-MLLMs, such as SkyEyeGPT [52], EarthDial [40], and EarthGPT [54], have shown promise in bridging satellite imagery and natural language for captioning, VQA, and scene grounding. However, most of these models rely on dense Transformer backbones that share parameters across all tasks and modalities, which limits specialization and leads to knowledge entanglement. While MoE models [5, 22] such as RingMoE [3] provide a pathway to task decomposition via parameter sparsity, their expert selection remains purely data-driven and lacks structural constraints, resulting in expert collapse and limited modularity [13, 14]. Moreover, RS-MLLMs rarely incorporate CoT prompting [31, 36, 55], despite its proven benefits in reasoning depth and interpretability. The absence of CoT impedes intermediate reasoning traceability, which is especially critical in high-stakes applications such as disaster response and environmental monitoring. To address these limitations, we propose GeoCoT, a manifold-constrained MoE framework. It promotes specialization via low-rank expert outputs and introduces CoT-driven reinforcement learning for interpretable and adaptive reasoning, handling diverse RS tasks with improved control and robustness.

3. Remote Sensing Dataset

To enhance the generalization and fine-grained reasoning in remote sensing, we construct high-quality datasets: Continual Pre-Training Dataset for pretraining and *RS-CoT-20k* Dataset for complex multi entity scene understanding.

3.1. Continual Pre-Training Dataset

Continual Pre-Training (CPT) aims to strengthen the base model ability to generalize and transfer across RS tasks. Prioritizing quality over quantity, we curate and organize

high quality image-text pairs from 8 open-source datasets for pre-training. We introduce the Data Distillation Regularization (DDR) method to filter and align the data, a process comprising the following steps: (1) Filter out text descriptions under 50 words to retain richer semantics. (2) Apply MinHash and LSH [53] to detect and remove duplicate texts, ensuring data diversity with 15-gram tokenization and a similarity threshold of 0.75. (3) Employ T5-base to expand incomplete text into well-structured and informative descriptions. (4) Remove images with an aspect ratio greater than 5 to avoid overly elongated features. (5) Employ RemoteCLIP [33] to compute image-text embeddings and remove pairs with similarity below 0.3.

In this way, the average text length reaches 667.12 characters, which significantly exceeds the existing RS datasets.

3.2. RS-CoT-20k Dataset

While CoT prompting has achieved success in various domains like math reasoning and code explanation, directly migrating it to RS tasks often fails to achieve satisfactory results. The understanding of RS images needs a progressive process from global to local. To solve this problem, we construct *RS-CoT-20k* dataset for RS-MLLMs including task-specific CoT prompts. Each prompt leads the model through global scene understanding and fine-grained visual analysis, ultimately yielding structured outputs for the task.

We collect multi-resolution RS images from open source datasets covering five tasks in Figure 1. The curated data spans 24 semantically rich RS scene categories with resolutions from 256×256 to 2519×2285 pixels. The dataset construction is guided by the following design criteria. Each data sample is formatted in a standardized manner and covers both common and uncommon RS scenes. We employ a Retrieval Augmented Generation (RAG) pipeline with GPT-4o [25], where external domain knowledge is retrieved based on task-specific prompts and fed into the model to generate CoT annotations tailored to various RS tasks. To further improve robustness and mitigate hallucinations, a portion of negative samples is included. The generated data are manually validated by domain experts in RS.

4. Method

4.1. Manifold Mixture-of-Experts

Unlike the unstructured high-dimensional distribution characteristics of natural images, RS images exhibit highly structured low-dimensional topological properties. These properties are predominantly reflected in large homogeneous regions, repetitive textures, and sparsely distributed small targets. These characteristics lead to substantial redundancy and suggest that most RS information resides on low-dimensional manifolds embedded in high-dimensional space. Motivated by these observations, we propose

Mani-MoE architecture (Figure 2) that integrates manifold-awareness into expert specialization.

Specifically, we introduce a rule-based local manifold mapping strategy to project high-dimensional RS representations into low-rank subspaces. This process eliminates the structural redundancy inherent in RS visual tokens while preserving task-relevant heterogeneous information. Given a token-level feature matrix $\mathbf{X} = [\mathbf{h}_1, \dots, \mathbf{h}_{L_v}]^\top \in \mathbb{R}^{L_v \times d}$, we first follow the classical graph-regularization paradigm to explicitly model the geometric affinities among visual tokens through their feature similarity, formulated as:

$$W_{ij} = \begin{cases} \exp\left(-\frac{\|\mathbf{h}_i - \mathbf{h}_j\|_2^2}{\varepsilon^2}\right), & \|\mathbf{h}_i - \mathbf{h}_j\|_2 \leq m, \\ 0, & \text{otherwise,} \end{cases} \quad (1)$$

$$m = \frac{1}{L_v(L_v - 1)} \sum_{i=1}^{L_v} \sum_{j=1}^{L_v} \|\mathbf{h}_i - \mathbf{h}_j\|_2, \quad i \neq j.$$

The weighted adjacency matrix $\mathbf{W}$ is employed to characterize the feature similarity among visual tokens, W_{ij} denotes the similarity between the i -th and j -th tokens. The mean distance of visual tokens is adopted as the threshold. Then, to prevent feature loss from uniform dimensionality reduction across the entire image, we partition the image and apply adaptive dimensionality reduction to different regions. Spectral clustering is performed to partition visual tokens into semantically consistent and geometrically coherent clusters. Based on the weight matrix $\mathbf{W}$, we first construct the normalized graph Laplacian:

$$\mathbf{L}_{\text{sym}} = \mathbf{I} - \mathbf{D}^{-1/2} \mathbf{W} \mathbf{D}^{-1/2},$$

$$\mathbf{D} = \text{diag}\left(\sum_{j=1}^{L_v} W_{1j}, \dots, \sum_{j=1}^{L_v} W_{L_v j}\right). \quad (2)$$

The normalized graph Laplacian $\mathbf{L}_{\text{sym}}$ preserves local structural properties and favorable spectral characteristics, where $\mathbf{D}$ denotes the degree matrix. We compute its eigenvalues $\{\lambda_k\}$, sorted in ascending order, and apply the eigen-gap heuristic to determine the number of clusters:

$$K = \arg \max_k (\lambda_{k+1} - \lambda_k). \quad (3)$$

The smallest K eigenvectors of $\mathbf{L}_{\text{sym}}$ are stacked and row-wise normalized to obtain the spectral embedding:

$$\mathbf{L}_{\text{sym}} \mathbf{u}_k = \lambda_k \mathbf{u}_k, \quad k = 1, \dots, K,$$

$$\mathbf{z}_{i,:} = \frac{[\mathbf{u}_1, \dots, \mathbf{u}_K]_{i,:}}{\|[\mathbf{u}_1, \dots, \mathbf{u}_K]_{i,:}\|_2}, \quad i = 1, \dots, L_v. \quad (4)$$

$\mathbf{Z} \in \mathbb{R}^{L_v \times K}$ represents the coordinates of all visual tokens in the low-dimensional spectral space. A standard clustering is applied to obtain the token partitions:

$$\{\mathcal{C}_1, \dots, \mathcal{C}_K\} = kmeans\left(\{\mathbf{z}_{i,:}\}_{i=1}^{L_v}, K\right). \quad (5)$$

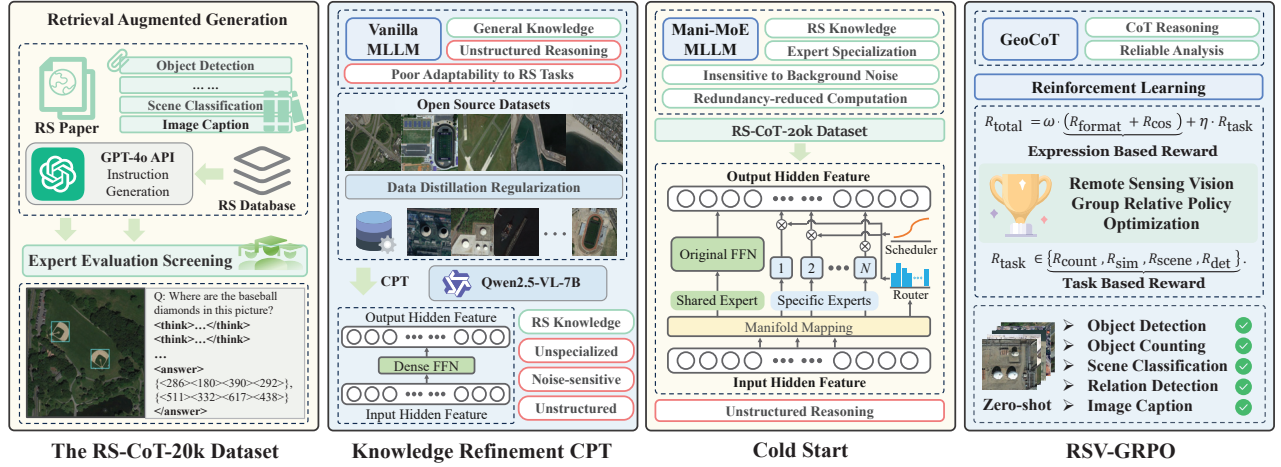


Figure 2. The sequential pipeline for developing GeoCoT. The *RS-CoT-20k* dataset is built with GPT-4o and retrieval-augmented generation for multi-task supervision. Knowledge refinement CPT aligns the base model with RS knowledge, while the cold start stage and Mani-MoE architecture enable expert specialization and noise-robust representations. RSV-GRPO strengthens global-to-target reasoning.

Denote by $\mathcal{C}_c$ the c -th token cluster obtained from spectral clustering, consisting of n_c visual tokens, expressed as: $\mathbf{X}_c = [\mathbf{h}_{c,1}, \mathbf{h}_{c,2}, \dots, \mathbf{h}_{c,n_c}]^\top \in \mathbb{R}^{n_c \times d}$. A spectral decomposition is then performed within each cluster:

$$\mathbf{X}_c = \mathbf{P}_c \mathbf{\Lambda}_c \mathbf{Q}_c^\top, \quad (6)$$

where $\mathbf{P}_c \in \mathbb{R}^{n_c \times v}$ and $\mathbf{Q}_c \in \mathbb{R}^{v \times d}$ are orthogonal matrices, $\mathbf{\Lambda}_c = \text{diag}(s_{c,1}, \dots, s_{c,r}, \dots, s_{c,v})$ contains non-negative spectral values sorted in descending order and $v = \min(n_c, d)$ denotes the maximum possible rank of $\mathbf{X}_c$. To retain the dominant structural components, only the top- r components with a cumulative energy ratio above the threshold ξ are preserved, where ξ is set to 0.85:

$$\frac{\sum_{t=1}^r s_{c,t}^2}{\sum_{t=1}^v s_{c,t}^2} \geq \xi. \quad (7)$$

The cluster-specific compressed representation is obtained as: $\hat{\mathbf{X}}_c = \mathbf{P}_c^{(r)} \mathbf{\Lambda}_c^{(r)} (\mathbf{Q}_c^{(r)})^\top \in \mathbb{R}^{n_c \times d}$. All clusters are then concatenated in order for following processing.

We adopt Qwen2.5-VL-7B [1] as the backbone. To improve structural efficiency and task specialization, we introduce our Mani-MoE modification: every three Transformer layers, the original MLP is replaced by a layer consisting of eight manifold experts and one shared expert. This modification increases the model size to 12B parameters. The shared expert is initialized with the original Qwen2.5-VL-7B weights to retain global scene understanding under complex background contexts. In contrast, the manifold experts operate in low-dimensional subspaces, enhancing object-centric reasoning and improving recognition accuracy in spatially dense remote sensing scenes. The token-to-expert

assignment in each routing layer is handled using the Top-2 routing strategy adopted from Switch Transformer [15]. Each token x is scored by a noisy gating function:

$$g = \text{softmax}(xW_g + \mathcal{N}(0, \sigma^2)), \quad (8)$$

W_g is a learnable gating matrix and Gaussian noise encourages routing diversity. The top-2 manifold experts with the highest scores process the token, and their outputs are aggregated based on the gating weights. An auxiliary load-balance loss is applied to prevent expert over-concentration.

4.2. Multi-stage Training Process

The multi-stage training strategy employed to optimize GeoCoT mainly consists of three key stages: (1) knowledge refinement CPT, (2) multi-task CoT cold start, and (3) RS Vision Group Relative Policy Optimization. Each stage progressively enhances multimodal reasoning capacity, task specialization, and overall generalization performance.

4.2.1. Knowledge Refinement CPT

To bridge the domain gap between natural and remote sensing images, a CPT stage is performed on a curated corpus of image-text pairs. This stage adapts the model's visual and textual representations to domain-specific patterns, substantially improving cross-modal alignment and transferability. This equips GeoCoT with domain-adapted reasoning capabilities, laying a solid basis for instruction tuning.

4.2.2. Multi-task CoT Cold Start

The instruction tuning is conducted under a cold-start setting, during which the Mani-MoE architecture is integrated into the pretrained backbone. The instruction tuning corpus consists of manually curated image-instruction-answer

triplets. CoT supervision is applied to guide intermediate reasoning steps, enable task-aware specialization and prepare for subsequent reinforcement learning alignment. To ensure stable training, cosine ramp-up schedule is used to gradually increase the gating weights of manifold experts. Each manifold expert is initialized with weights sampled from Gaussian distribution $\mathcal{N}_s(\mu, \sigma^2)$, where μ and σ^2 are estimated from the parameter statistics of shared expert. Additionally, Low-Rank Adaptation (LoRA) is applied only to attention modules without full parameter updates.

4.3. RS Vision Group Relative Policy Optimization

We develop a task-specific RS Vision Group Relative Policy Optimization (RSV-GRPO) framework for various remote sensing tasks to guide the policy optimization.

4.3.1. Preliminary

Group Relative Policy Optimization (GRPO) [18] normalizes grouped advantages via reward comparison to directly approximate the advantage. It first generates M distinct responses $\{o_i\}_{i=1}^M$ sampled for a query q , each response receives a feedback $\{g_i\}_{i=1}^M$ derived from task-specific reward signal. Define the i -th response's relative advantage as:

$$A_i = \frac{g_i - \text{mean}(\{g_1, \dots, g_M\})}{\text{std}(\{g_1, \dots, g_M\})}. \quad (9)$$

To update the policy π_θ , incorporating both the normalized advantage and a regularization penalty against a reference policy π_{ref} , yielding the following training objective:

$$\begin{aligned} \mathcal{J}_{GRPO}(\theta) = \mathbb{E} [q \sim \mathbb{P}(Q), \{o_i\}_{i=1}^M \sim \pi_{\theta_{\text{old}}}(O|q)] \\ \frac{1}{M} \sum_{i=1}^M \frac{1}{|O_i|} \sum_{t=1}^{|O_i|} \left\{ \min[\rho_i \cdot A_i, \right. \\ \left. \text{clip}(\rho_i, 1 - \epsilon, 1 + \epsilon) \cdot A_i] \right. \\ \left. - \beta \mathbb{D}_{KL}[\pi_\theta \parallel \pi_{\text{ref}}] \right\}, \end{aligned} \quad (10)$$

where $\rho_i = \frac{\pi_\theta(o_i|q)}{\pi_{\text{old}}(o_i|q)}$ is the policy ratio, and ϵ and β are used to control the clipping range and KL penalty strength.

4.3.2. Reward Design for Multiple RS Tasks

To supervise MLLM learning on diverse RS tasks, we propose a reward aggregation strategy:

$$\begin{aligned} R_{\text{total}} = \omega \cdot \underbrace{(R_{\text{format}} + R_{\text{cos}})}_{\text{Expression Based Reward}} + \eta \cdot R_{\text{task}}, \\ R_{\text{task}} \in \underbrace{\{R_{\text{count}}, R_{\text{sim}}, R_{\text{scene}}, R_{\text{det}}\}}_{\text{Task Based Reward}}. \end{aligned} \quad (11)$$

The overall reward function consists of two components: an expression based reward and a task-specific reward. We assign different weights to each component of the reward.

Format Reward. The format based reward function requires the model to reason and answer in the following format: `<think> reasoning content </think>`, `<answer> final answer </answer>`. The reward function checks each completion c_i for conformity to the above format. $R_{\text{format}}(c_i) = 1.0$ if c_i matches the required format.

Accuracy Reward. For counting tasks, we define the reward by comparing the predicted count $\hat{y}_i$ with the ground truth count y_i . If and only if $\hat{y}_i = y_i$, $R_{\text{count}}(i) = 1.0$.

For image caption and relation detection tasks, we assess semantic consistency between the model-generated answer $\hat{a}_i$ and the ground truth answer a_i using cosine similarity:

$$R_{\text{sim}}(i) = \cos(f(\hat{a}_i), f(a_i)), \cos(f(\hat{a}_i), f(a_i)) \geq \tau. \quad (12)$$

Here, $f(\cdot)$ denotes the sentence encoder, for which we use MEXMA. τ is a fixed similarity threshold. The cosine similarity is calculated by: $\cos(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \cdot \|\mathbf{v}\|}$.

For scene classification task, we compute the reward based on the consistency between the predicted label set $\hat{Y}_i$ and the ground truth label set Y_i . We introduce F1 score for the cases where partial predictions are correct:

$$R_{\text{scene}}(i) = F_1(\hat{Y}_i, Y_i). \quad (13)$$

The F1 score is calculated by the following method:

$$F_1(\hat{Y}_i, Y_i) = \frac{2 \cdot |\hat{Y}_i \cap Y_i|}{|\hat{Y}_i| + |Y_i|}. \quad (14)$$

Object Detection Reward. In detection task, we propose a reward function that evaluates both IoU metric and confidences, perform the normalization operation:

$$R_{\text{det}}(i) = \alpha \cdot (R_{\text{iou}}(i) + R_{\text{conf}}(i)). \quad (15)$$

Cosine Reward. This reward mechanism promotes concise answers when the model is confident, while encouraging more elaborative reasoning under uncertainty. This reward is computed using the following function:

$$R_{\text{cos}}(t) = \delta - (\delta - \gamma) \cdot \frac{1 - \cos(\frac{t\pi}{T})}{2}, \quad (16)$$

where t is the token length of the generated output, T is the maximum length threshold, γ and δ are the reward bounds, chosen based on prediction correctness:

$$(\gamma, \delta) = \begin{cases} (\gamma_c, \delta_c), & \text{the answer is correct,} \\ (\gamma_w, \delta_w), & \text{the answer is incorrect.} \end{cases} \quad (17)$$

The cosine schedule avoids abrupt changes by smoothly interpolating reward values within the interval $[\gamma, \delta]$, controlled by the length t . We set: $(\gamma_c, \delta_c) = (1.0, 0.5)$ for correct answers, $(\gamma_w, \delta_w) = (-0.5, 0.0)$ for incorrect answers, $T = 1024$ as the maximum token length.

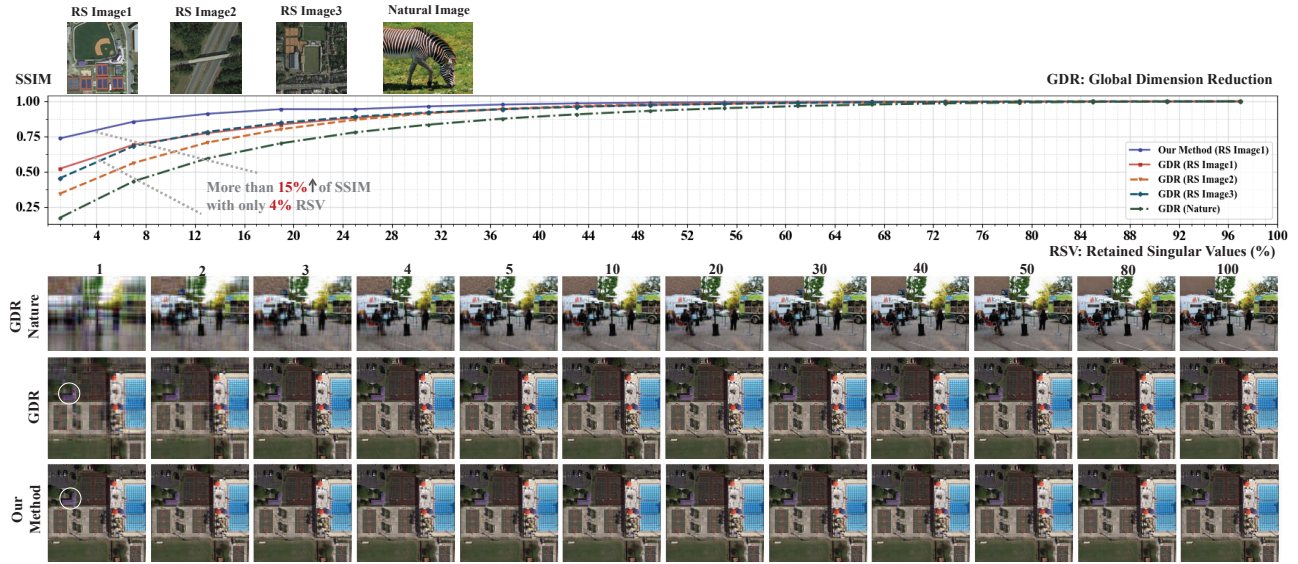


Figure 3. Comparison between Global Dimension Reduction and the Proposed Local Manifold Mapping Strategy.

Table 1. Comparison of image caption Benchmark on UCM-Captions dataset. **Bold** represents the best performance per training strategy.

Model	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr
<i>Specialist Models: RS customized VLMs</i>							
RSGPT	86.12	79.14	72.31	65.74	42.21	78.34	333.23
GeoChat	79.94	72.52	65.83	59.32	37.66	74.80	201.86
SkySenseGPT	81.21	74.69	67.56	61.41	40.24	76.49	236.19
<i>Generalist Models: open-world multimodal LLMs</i>							
Llama 4	47.92	39.13	30.02	21.75	18.90	37.65	39.65
GPT-4o	54.05	46.52	36.15	30.87	25.28	45.45	54.43
Gemini 2.5	48.98	42.30	31.25	24.12	21.77	39.29	45.69
GeoCoT	86.31	80.15	72.79	67.14	40.10	80.18	202.38

5. Experiment

5.1. Implementation Details

The MLLM backbone model in our GeoCoT framework is partially initialized based on the publicly released Qwen2.5-VL-7B checkpoint. The training process includes three consecutive stages: CPT, cold start and RSV-GRPO. In the first stage, we fine-tune the model on our CPT Dataset for 3 epochs with batch size 2. In the second stage, we replace the original FFN layers with sparse Mani-MoE modules, perform cold-start training on five remote sensing tasks. This process involves CoT reasoning. The training employs LoRA with a rank of 32 for 3 epochs and a batch size of 4. Finally, we employ RSV-GRPO to refine the output distribution, enhancing task-specific alignment and generalization capabilities. All training is performed on 8 NVIDIA A100 GPUs (80GB).

5.2. Comparison

5.2.1. RS Multimodal Interaction

To demonstrate the practical capabilities of GeoCoT, we construct a set of interactive examples that simulate typical vision-language usage scenarios in RS. Representative cases and qualitative comparisons are shown in Figure 4. Compared to GPT-4o and Qwen2.5-VL, GeoCoT exhibits a higher degree of instruction alignment and generates more accurate responses for RS spatial reasoning tasks, highlighting its effectiveness in complex multimodal RS queries.

5.2.2. Main Results

To comprehensively assess the performance of GeoCoT across various RS tasks, we adopt standard benchmarks for each task: AID for scene classification, RSVQA_LR for object counting, DIOR for both object detection, UCM-Captions for image caption, and DOTA-v2.0 for relation de-

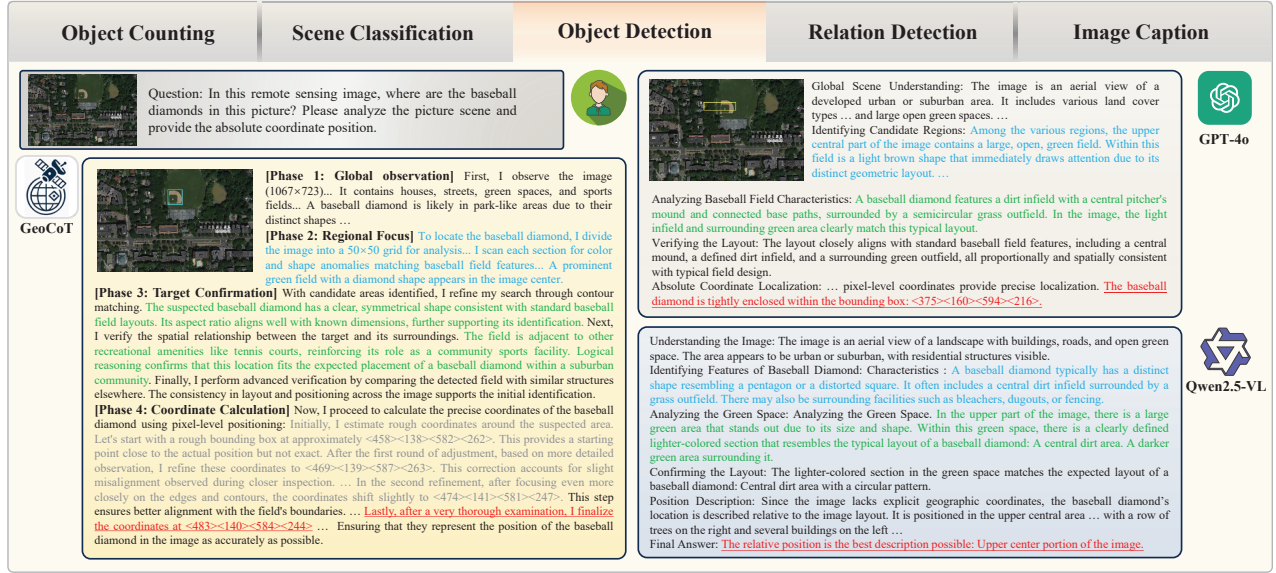


Figure 4. Representative VQA examples illustrating the strong performance of GeoCoT.

Table 2. Comparison of 4 RS tasks across models. Each column reports the most relevant evaluation metric on the corresponding dataset. “@0.5” and “@[.5:.95]” refer to mAP@0.5 and mAP@[.5:.95], respectively. Underlined results denote zero-shot performance.

Model	Relation Detection DOTA-v2.0 (F1-Score)	Scene Classification AID (Acc.)	Object Counting RSVQA_LR (Acc.)	Object Detection	
				DIOR (@0.5)	DIOR (@[.5:.95])
<i>Small Models: classical vision-only methods</i>					
Faster R-CNN [38]	N/A	N/A	N/A	76.50	41.43
DETR [8]	N/A	N/A	N/A	73.12	47.87
<i>Specialist Models: RS customized VLMs</i>					
RSGPT [23]	45.93	<u>85.53</u>	3.84	59.02	33.60
GeoChat [26]	N/A	<u>72.03</u>	5.74	62.40	36.83
SkySenseGPT [34]	74.34	<u>92.25</u>	9.20	69.94	48.17
<i>Generalist Models: open-world multimodal LLMs</i>					
Llama 4	<u>32.22</u>	<u>53.20</u>	<u>5.85</u>	<u>49.32</u>	<u>27.66</u>
GPT-4o	<u>41.83</u>	<u>58.81</u>	<u>7.34</u>	<u>54.56</u>	<u>30.57</u>
Gemini 2.5 [41]	<u>44.05</u>	<u>57.45</u>	<u>6.02</u>	<u>51.49</u>	<u>28.91</u>
GeoCoT	81.21	91.73	17.15	78.29	54.95

tection. Specialist models are designed specifically for RS scenarios, while generalist models target a broad spectrum of cross-domain tasks. We report the results of 4 tasks in Table 2, and present a separate benchmark for image caption in Table 1 using metrics specific to language generation.

A comparative analysis of model performance on 4 representative RS tasks is provided in Table 2. Classical models like Faster R-CNN perform reasonably in detection. RS-specific models such as RSGPT exhibit stronger task alignment, with SkySenseGPT achieving the highest scene classification accuracy. Generalist models including Llama 4 Scout, GPT-4o, and Gemini 2.5 Flash demonstrate moderate zero-shot transferability but consistently underperform compared to specialized counterparts. GeoCoT attains the

highest overall performance across all 4 tasks. It is important to emphasize that, despite the amazing accuracy of previous RS-MLLMs on specific tasks, such overfitting training often resulted in severe hallucinations outside of the training content. GeoCoT does not consider test accuracy as the only reference, it places greater emphasis on the generation of reliable and semantically faithful responses.

We benchmark the image caption capabilities of various models using the UCM-Captions dataset, with results summarized in Table 1. Performance is measured using a suite of standard captioning metrics, including BLEU, METEOR, ROUGE_L and CIDEr. Among specialist models, SkySenseGPT performs well in terms of linguistic quality and lexical coverage, while generalist models such as

Table 3. Ablation study on model training pipeline across 5 RS tasks. “GeoCoT” includes CPT, cold start, and RSV-GRPO. Here, ① and ② denote the configurations without and with the local manifold mapping strategy, respectively.

Configuration	Scene Classification <i>AID (Acc.)</i>	Object Counting <i>RSVQA_LR (Acc.)</i>	Object Detection <i>DIOR (mAP@[.5:.95])</i>	Relation Detection <i>DOTA-v2.0 (F1-Score)</i>	Image caption <i>UCM-Captions (ROUGE_L)</i>
Baseline	69.42	5.21	41.22	65.34	65.26
+CPT	75.81	8.56	45.61	72.76	71.41
①+Cold Start(Classic-MoE)	82.28	13.25	49.36	74.90	72.24
②+Cold Start(Mani-MoE)	87.06	15.07	51.71	77.82	75.49
GeoCoT(Mani-MoE)	91.73	17.15	54.95	81.21	80.18

Table 4. Effect of MoE expert quantity on GeoCoT performance with the same backbone and routing strategy after cold start.

Experts (Params)	Scene Classification <i>AID (Acc.)</i>	Object Counting <i>RSVQA_LR (Acc.)</i>	Object Detection <i>DIOR (mAP@[.5:.95])</i>	Relation Detection <i>DOTA-v2.0 (F1-Score)</i>	Image Caption <i>UCM-Captions (ROUGE_L)</i>
2 (8.3B)	77.04	7.11	47.61	72.31	70.44
4 (9.5B)	80.82	11.81	49.24	74.79	73.21
8 (12.0B)	87.06	15.07	51.71	77.82	75.49
16 (17.1B)	84.91	13.01	51.59	76.84	77.13

GPT-4o and Gemini 2.5 exhibit certain zero-shot capabilities. GeoCoT consistently delivers the most competitive results across all metrics, underscoring its strength in regional understanding and fine-grained captioning for RS images.

5.2.3. Comparative Visualization of Manifold Strategy

To further demonstrate the low-dimensional topological characteristics of RS images that are different from natural images and validate the superior performance of our rule-based local manifold mapping strategy over conventional approaches in RS images dimensionality reduction and redundancy suppression, we visualize the comparison results in Figure 3. Our method attains significant SSIM gains with only a small number of average retained singular values.

5.2.4. Efficiency and Practical Deployability of GeoCoT

We evaluate GeoCoT on a single NVIDIA RTX 4090 GPU (24 GB). The 12B model loads and runs within 21.99 GB memory usage for inference. On a prompt of 360 input tokens, the model achieves approximately 22.56 tokens/s generation throughput and generates 536 tokens in 23.76 seconds. Even on a single NVIDIA A100 GPU (80 GB), GeoCoT reaches an inference speed of 23.53 tokens/s. These results demonstrate that GeoCoT is efficient and deployable.

5.3. Ablation

5.3.1. Ablation Study on Training Pipeline

To assess the impact of each component in the training pipeline, we conduct a stepwise ablation study that transitions from a standard dense architecture to the full MoE-enhanced GeoCoT model. Beginning with the pretrained Qwen2.5-VL-7B checkpoint, we first apply CPT to adapt the base model to RS distributions. Subsequently, we replace the FFN layers with sparse expert modules and perform cold start to align with multimodal instruction-

following tasks. In the final stage, we introduce RSV-GRPO to further refine the outputs using human-aligned reward signals. We report the most relevant validation metric for each task. The results, presented in Table 3, show consistent improvements at every stage, validating the effectiveness of our progressive optimization strategy from general-purpose initialization to domain-specialized expert modeling. Rows 3 to 4 in the ablation study indicate that the local manifold mapping strategy brings notable performance gains.

5.3.2. Impact of MoE Expert Quantity

We perform an ablation study to assess the effect of the number of experts in the MoE architecture. We evaluate models with 2, 4, 8, and 16 experts on 5 tasks, the results are summarized in Table 4. Increasing the number of experts generally improves performance, with 8 experts achieving the best results. Increasing to 16 experts yields limited gains, likely due to routing uncertainty and expert under-utilization. Meanwhile, we conduct an experiment showing that performance gains stem from expert specialization, not just an increase in parameters.

6. Conclusion

We innovatively build GeoCoT, an RS-MLLM designed to improve task-specific accuracy and reasoning capability in RS. Specifically, we propose a Mani-MoE architecture that projects high-dimensional data into a low-rank manifold subspace. This preserves intrinsic salient features while suppressing redundant attention and noise, significantly enhancing task-level performance. To optimize the above architecture, we introduce a CoT-driven multi-stage training strategy, remarkably augmenting the reasoning ability of models. Furthermore, we construct *RS-CoT-20k* dataset to remedy the lack of structured CoT supervision in RS. Extensive experiments show the superiority of GeoCoT.

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